

ADETORO DAMILOLA SIMBIAT

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SUMMARY

A masters student of Engineering Artificial Intelligence with interest in the integration of Artificial Intelligence to build smart softwares. My research interests lie in the application of artificial intelligence to healthcare.

EDUCATION

Carnegie Mellon University — *Masters in Engineering Artificial Intelligence* CGPA 3.86/4.00 August 2025 - May 2027

Federal University of Technology, Minna — *Bachelor of Technology* November 2018 - Sept 2024
Electrical Electronics Engineering , CGPA 4.5/5.0 (Top 5%)

Spark Academy — *Deep learning and medical imagery* February 2026- August 2026

PROFESSIONAL EXPERIENCE

Check-Me — AI Specialist

May 2026 - Present

- Researched and developed ML models for breast health risk assessment, improved the model interpretation for breast cancer scans using a novel approach.
- Built scalable data pipelines, evaluated model performance, and implemented optimisations to improve accuracy
- Collaborated with engineering and research teams to integrate AI models into backend services and mobile applications

AEIRG, Minna — Software intern

Oct 2023 - March 2024

- Develop the backend architecture for the smart aquaculture system.
- Create scalable and efficient systems to manage data from various sensors, monitoring devices, and control systems.
- Collaborate with data scientists and researchers to implement algorithms for data analysis, predictive modeling, and decision support systems within the aquaculture domain.
- Assisted in the development of AI/ML models for pipeline fault detection in south-eastern part of Nigeria

AI4CE, Minna — Research Assistant

May 2020- January 2021

- Collaborated with a senior researcher to explore renewable energy options aimed at reducing environmental pollution and promoting sustainable energy.
- Assisted in the preparation and submission of grant proposals to secure funding for renewable energy research projects, and collaborated with team members to manage project budgets and timelines.
- Played a significant role in the research project, contributing to 40% of the effort by assisting in the development and documentation of multiple viable clean energy options applicable in Minna, Niger State

TECHNICAL SKILLS

- **Computer languages:** Python, C++, Linux.
- **Web application tools:** FastAPI, Flask, Django, JavaScript, Docker, Kubernetes, Azure
- **Artificial Intelligence:** Regression, Classification, Clustering, Feature engineering, Anomaly Detection, Medical imaging

RESEARCH

Cardiovascular disease diagnosis in low resource settings

Supervisor: Prof. Carine Makumakuza

October 2025 - Present

Developed machine learning and deep learning models leveraging ECG and PCG signals to enable early detection and severity classification of rheumatic heart disease (aortic stenosis) and arrhythmias, targeting cost-effective diagnosis in medically underserved communities

FPN-Mamba: MRI Classification with Feature Pyramid Networks & Mamba State Space Models

Designed a hybrid architecture combining Feature Pyramid Networks with Bidirectional Mamba State Space Models for multi-scale MRI classification, achieving linear-time $O(L)$ complexity and outperforming the InceptionNet baseline on medulloblastoma screening across four of five metrics.

Deep Learning for Early Detection of Diabetic Retinopathy in Low-Resource Settings

Developed a resource-efficient deep learning model for early diabetic retinopathy detection from retinal fundus images, optimized for deployment on low-cost edge devices in low-resource healthcare settings

AWARDS AND ACHIEVEMENTS

Deep learning Indaba 2026 Reviewer, MINDS Scholarship, Dean's list, (Awarded to students with the top 90 percentile grades), Funmi Omo grant and mentorship program, PTDF Scholarship, Grace Hopper Conference Scholarship, UNIDO KPABEP Scholarship, Project support lead for ingressive for good(Futminna chapter)